

Aryan Bhavsar

Machine Learning Systems Engineer | Forward-Deployed AI & Full-Stack

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Ahmedabad, India

LinkedIn

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LeetCode

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science Engineering; CGPA: 8.65

2021 – 2025

Chennai, India

Zydus School for Excellence

Higher Secondary Education (12th Grade: 84%; 10th Grade: 89%)

2019 – 2021

Ahmedabad, India

EXPERIENCE

Hitech Digital Solutions

Sept 2025 – Present

Ahmedabad, India

Machine Learning Systems Engineer (Forward Deployed AI & Full Stack)

Mar 2026 – Present

- Architected and built an in-house real-estate document extraction and classification platform end-to-end (React/Vite frontend, FastAPI backend, SQLite job queue) within a few months, serving as primary ML/full-stack engineer on a 4-person team.
- Replaced a paid Azure OCR service and a legacy BERT classifier with fine-tuned open-source OCR engines (DocTR, Donut, Monkey OCR) and LLMs/VLMs (Qwen 2/2.5/3, Mistral, InternVL) using LoRA fine-tuning via Unsloth, reaching 90%+ extraction accuracy at zero inference cost on in-house GPUs.
- Engineered a vLLM-based inference layer with sidecar health monitoring and zero-downtime model hot-swapping, cutting per-document processing time (deed: 15s→1s, mortgage: 25s→6s, classification: 5s→0.5s) while scaling to 50,000+ documents/day in production.
- Built a custom confidence-scoring system to route low-confidence predictions to manual review, cutting manual verification headcount from 500 to under 100 and contributing to a 33% reduction in operational costs and a 10% profit increase last quarter; deployed the platform via RunPod and Docker for in-house and external users.
- Star Performer of the Month & Quarter** — certified internal award for leading the document-intelligence platform.

Graduate Engineer Trainee (Jr. Data Scientist)

Sept 2025 – Feb 2026

- Built an Intelligent Pricing & Changelog Management System, benchmarking open-source models for multiple services and functionalities.
- Applied GPT-4o Vision with Chain-of-Thought (CoT) prompting on complex menu structures and integrated Hybrid Reconciliation via Sentence-Transformers to track price elasticity and automate data entry with Pydantic schemas.

Hyperface Technologies Pvt. Ltd.

Feb 2025 – Aug 2025

AI Researcher (Generative AI Team)

Bengaluru, India

- Architected a Multi-Agent System using FastAPI integrated with LLMs (GPT-4, Gemini) to drive intelligent workflow automation.
- Engineered agent-level prompt strategies and pipeline logic, optimizing response accuracy and reducing latency.
- Developed a robust evaluation framework using Hugging Face open-source models to benchmark LLM performance.

Indian Space Research Organisation (ISRO)

Oct 2024 – Jan 2025

Machine Learning Intern

Ahmedabad, India

- Designed a deep learning classifier to distinguish snow vs. cloud cover in satellite imagery utilizing custom CNN architectures.
- Enhanced crop yield prediction accuracy by aggregating multi-spectral image data and analyzing temporal patterns.

PROJECTS

Multi-Agent Real Estate Copilot | Python, FastAPI, Gemini, Docker

- Built a multi-agent backend with FastAPI to route queries between a vision-capable Property Agent and a text-based Tenancy Agent.
- Leveraged Google Gemini Pro Vision API for real-time analysis of property images, integrating insights into a Next.js chat interface.

Virtual Football Analysis | YOLOv8, OpenCV, Flask, CNN

- Implemented YOLOv8 for real-time player and ball tracking, deploying the computer vision pipeline via a Flask web application.
- Automated the generation of event-based match highlights using time-indexed object detection.

PUBLICATIONS

Deep Learning for Tea Leaf Disease Synthesis & Detection – ScienceDirect

Markerless Data-based Cyclist Posture Detection using Deep Learning – IEEE Xplore

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript, R, HTML, CSS

Frameworks: PyTorch, TensorFlow, FastAPI, Django, Flask

GenAI & NLP: LangChain, RAG, CAG, Qwen, Monkey OCR, DoNut, GPT-4o, Multi-modal Agents

Cloud & DevOps: GCP, Docker, Git, GitHub & GitLab Actions

Databases: MySQL, SQLite, PostgreSQL

Libraries: Spacy, Fuzzywuzzy, Beautiful Soup, Playwright, OpenCV, Pandas, NumPy, Scikit-learn